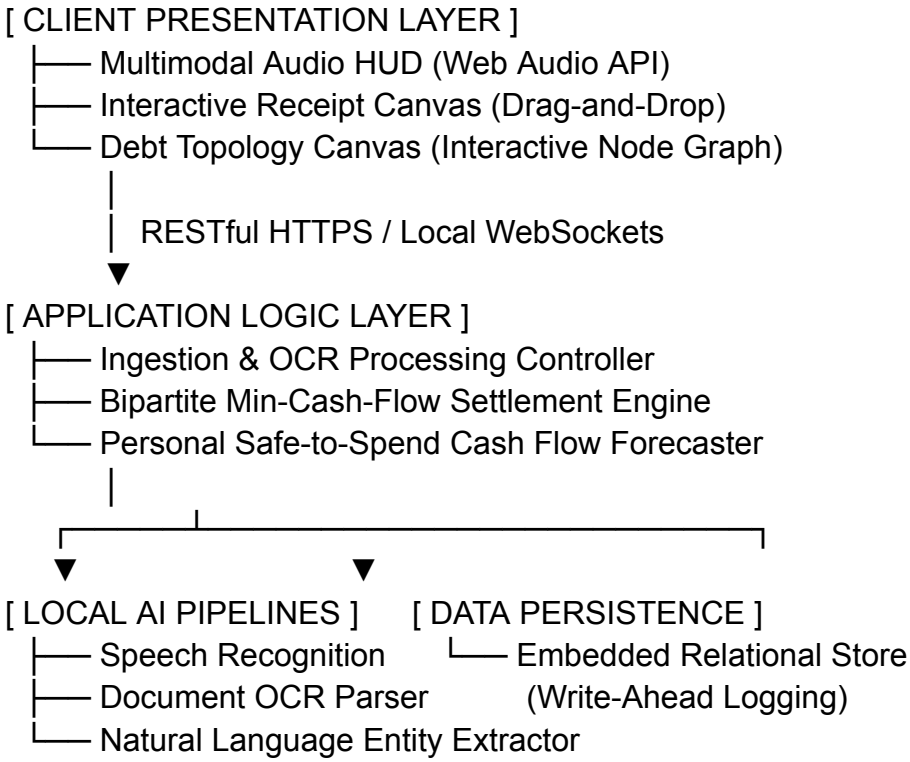
High-Level Design (HLD)

ClarifyFinance: System Architecture & Interaction Design

1. System Architecture & Topology

ClarifyFinance is structured as a modular full-stack architecture. The presentation tier manages visual states, multimodal inputs, and direct-manipulation canvases. The application tier manages transactional logic, graph optimizations, and AI pipelines.



2. Subsystem Decomposition

2.1 Presentation Tier

- **Receipt Interaction Controller:** Displays OCR-extracted receipt items as discrete visual elements. Enables touch and mouse assignment where users associate group avatars with items, recalculating individual subtotals and proportional tax in real time.

- **Settlement Topology Viewer:** Renders group debt distributions as an interactive directed network. Nodes represent members with net balance indicators, while edges represent debt payment flows. Selecting an edge reveals an explanatory modal with full transaction provenance.
- **Multimodal Capture Bar:** Manages audio stream acquisition, displaying dynamic waveform activity to assure the user that voice input is active, and provides editable staging chips upon transcription.

2.2 Application Services Tier

- **Ingestion Gateway:** Accepts audio payloads and document images, routing them to local processing pipelines asynchronously without blocking user interactions.
- **Settlement Engine:** Ingests raw multi-party transaction histories, computes net balances, executes greedy bipartite transaction simplification, and synthesizes human-readable audit explanations for simplified paths.
- **Cash Flow Forecaster:** Evaluates individual transactions, recurring scheduled bills, and pending group credits to calculate the forward-looking Safe-to-Spend metric.

2.3 AI & Algorithmic Subsystems

- **Speech-to-Text Pipeline:** Uses quantized, local open-source speech models to transcribe audio inputs into plain text with word-level accuracy.
- **Document OCR Pipeline:** Detects textual regions and line groupings on uploaded receipts, matching item descriptions with corresponding prices.
- **Structured NLU Extractor:** Translates freeform transcripts and OCR lines into standardized entities (payers, item amounts, currencies, and split ratios).

3. Conceptual Data Model

The application uses an embedded relational data architecture configured for high local read concurrency and data durability.

Entity	Primary Attributes	Relationships	Core Purpose
User	User ID, Name, Email, Payment Handle	1-to-Many with Group Memberships, Expenses	Represents individual participants and system owners.

Group	Group ID, Name, Created Timestamp	1-to-Many with Expenses, Members	Encapsulates shared trips, events, or households.
Expense	Expense ID, Group ID, Payer ID, Total, Tax, Tip	Many-to-1 with Groups, 1-to-Many with Splits	Records primary financial outlays and receipts.
Receipt Item	Item ID, Expense ID, Name, Unit Price, Quantity	Many-to-1 with Expenses, Many-to-Many with Users	Stores individual line items parsed from receipts.
Expense Split	Split ID, Expense ID, Member ID, Assigned Amount, Status	Many-to-1 with Expenses, Many-to-1 with Users	Represents individual debt liabilities per transaction.
Settlement	Settlement ID, Debtor ID, Creditor ID, Amount, Status	Many-to-1 with Groups	Records resolved payments between participants.

4. Mathematical Formulations & Algorithmic Design

4.1 Proportional Itemization and Surcharge Allocation

When splitting bills, items may be shared by subsets of users while taxes and gratuities apply globally. Let M be the total number of items, where item k has price pk. Let Uk be the set of participants sharing item k. Each participant i's base subtotal bi is calculated as:

$$b_i = \sum_{k=1}^{M} \frac{p_k}{|U_k|} \quad \forall i \in U_k$$

With global tax T, gratuity S, and discounts D, the proportional overhead multiplier α is:

$$\alpha = \frac{T + S - D}{\sum_{k=1}^{M} p_k}$$

The final balance Ci owed by participant i is:

$$C_i = b_i \times (1 + \alpha)$$

This ensures the sum of all individual balances equals the total receipt amount exactly, down to the smallest currency unit.

4.2 Greedy Min-Cash-Flow with Audit Tracking

To minimize the number of payments required to settle group debts:

1. **Net Balance Calculation:** For each member, sum all amounts paid for others and subtract all amounts owed to others.
2. **Partitioning:** Separate members into net debtors (negative balance) and net creditors (positive balance).
3. **Greedy Matching:**
    - Identify the debtor with the largest debt and the creditor with the largest credit.
    - Settle an amount equal to the minimum of the two balances.
    - Create a simplified transaction between these two members.
    - Document the transitive path (e.g., Member A settles with Member C because Member A owed Member B, who in turn owed Member C).
    - Update balances and repeat until all net positions reach zero.

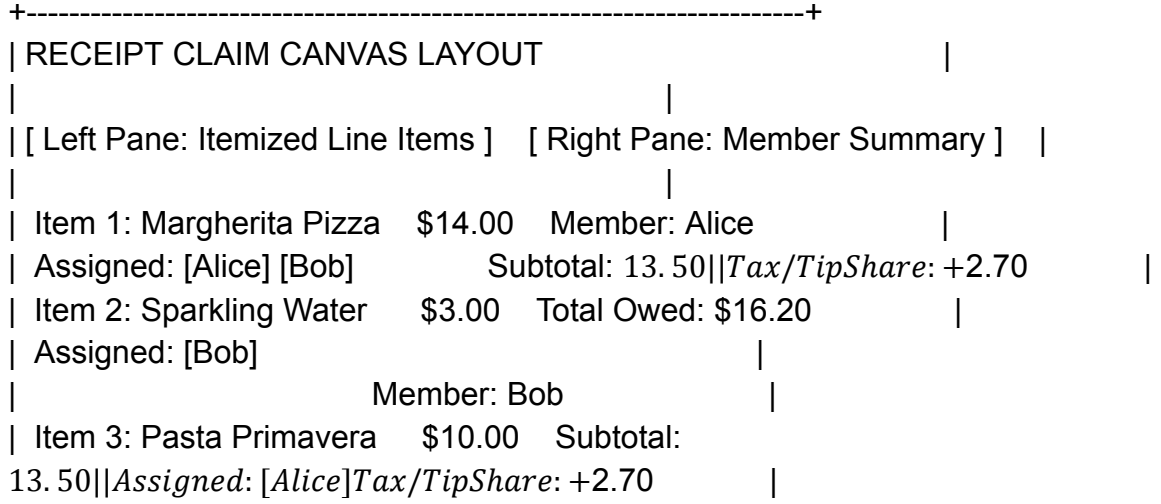
4.3 Safe-to-Spend Balance Model

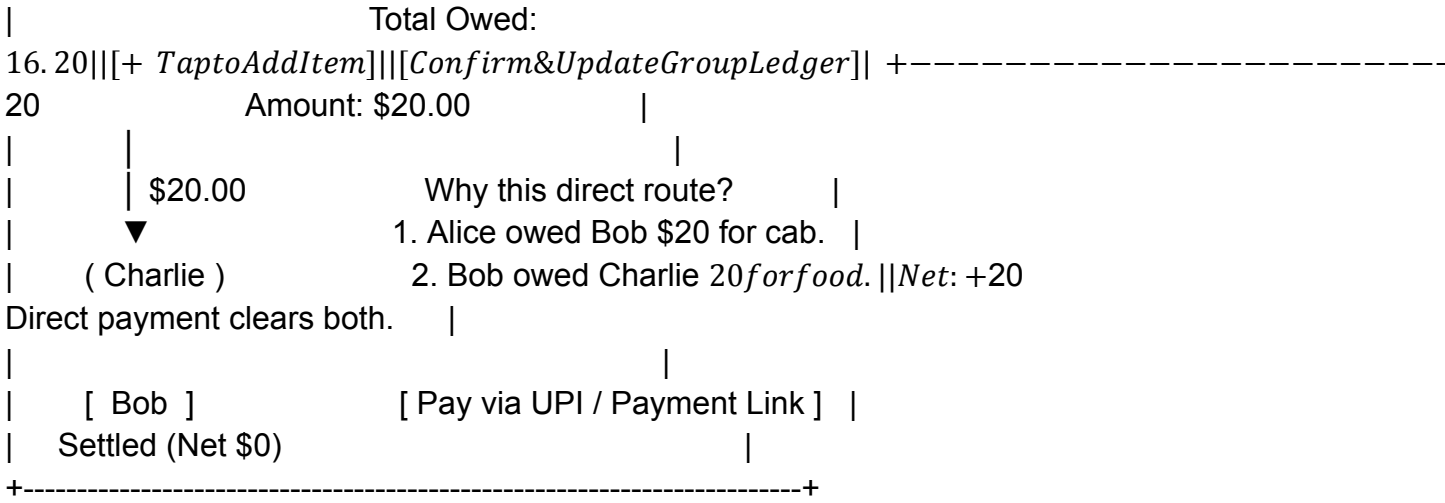
The personal budgeting view avoids backward-looking guilt by projecting forward discretionary headroom:

$$\text{Safe-to-Spend} = B_{\text{checking}} - \sum R_{\text{upcoming}} - \sum E_{\text{fixed}} + \sum (G_{\text{receivables}} \times \omega_u)$$

where B_{checking} is the liquid account balance, R_{upcoming} represents scheduled recurring bills, E_{fixed} represents estimated nondiscretionary living costs, and $\omega_u \in [0.0, 1.0]$ represents a reliability weight applied to group member u's pending repayment.

5. Interaction Design & Screen Architecture





6. Usability Evaluation Protocol for Academic Review

To evaluate the system under Human-Computer Interaction benchmarks, the following study methodology is established:

1. **Experimental Design:** A within-subjects comparative study (N=12 to 16 participants) evaluating ClarifyFinance against a standard baseline (conventional split app combined with manual spreadsheet calculations).
2. **Standardized Tasks:**
    - **Task 1:** Enter a multi-item restaurant receipt with shared appetizers, separate beverages, tax, and gratuity.
    - **Task 2:** Interpret a simplified debt settlement and identify why a specific payment path was assigned.
    - **Task 3:** Log an expense using voice input and correct a misidentified split via staging chips.
3. **Data Collection & Instruments:**
    - **Task Completion Time (Seconds):** Measured from initial input to final confirmation.
    - **System Usability Scale (SUS):** 10-item questionnaire administered immediately following each test block.
    - **NASA Task Load Index (NASA-TLX):** Assesses Mental Demand, Temporal Demand, and Frustration levels.
    - **Trust & Algorithmic Transparency Survey:** Assessed using a 7-point Likert scale focused on settlement clarity.